AnaeCare: An Explainable Multimodal Deep Learning Framework for Non Invasive Anemia Detection

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Problem Statement

- Traditional diagnosis of anemia requires invasive blood tests.
- Cost, diagnostic delays in rural and underserved areas.
- Early non invasive anemia detection protects vulnerable lives.

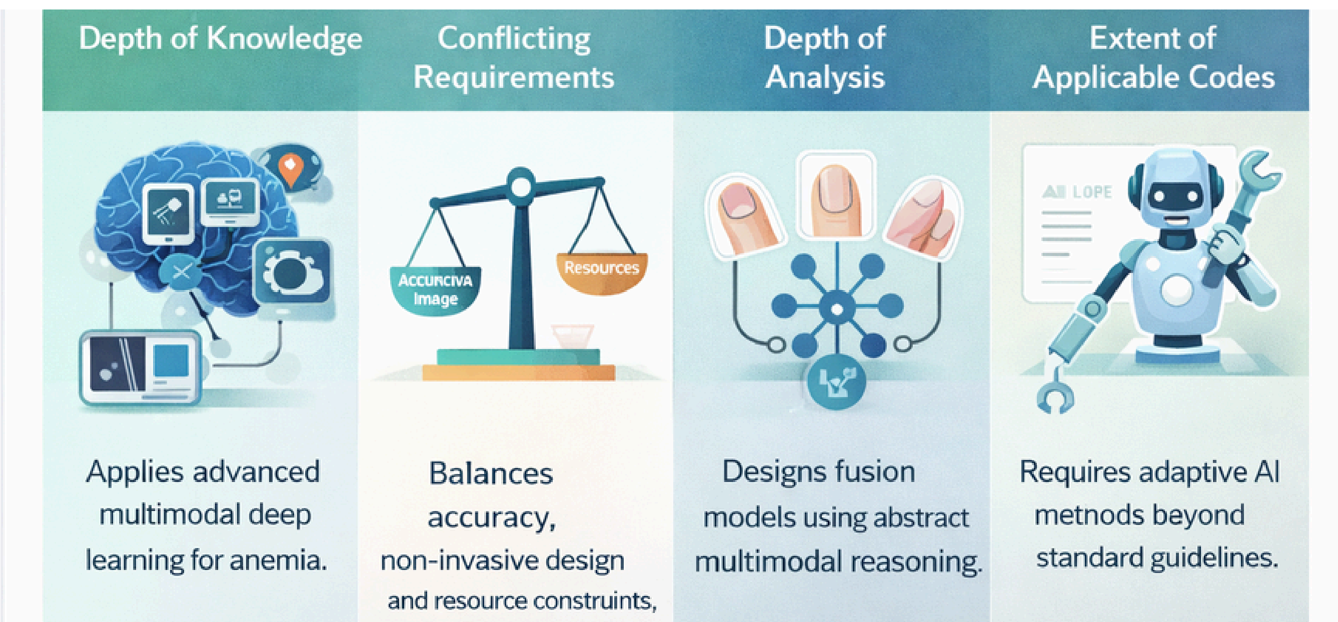
Research Questions

Can early detection save vulnerable lives?

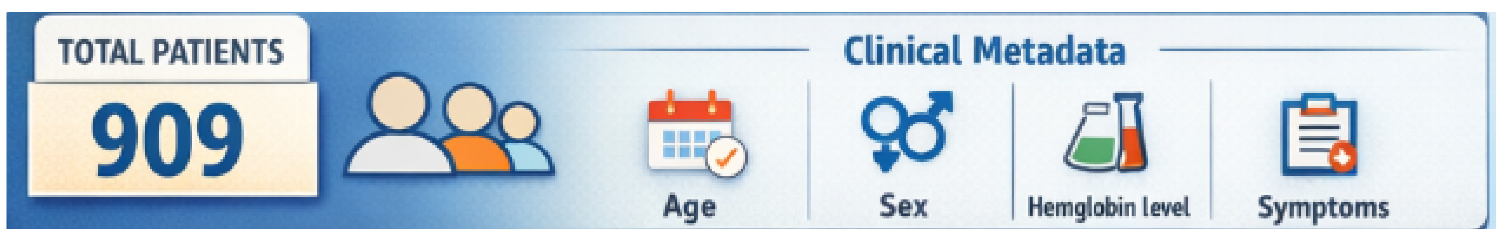
How accurate is noninvasive AI screening?

Does multimodal fusion stratergies outperform single source models?

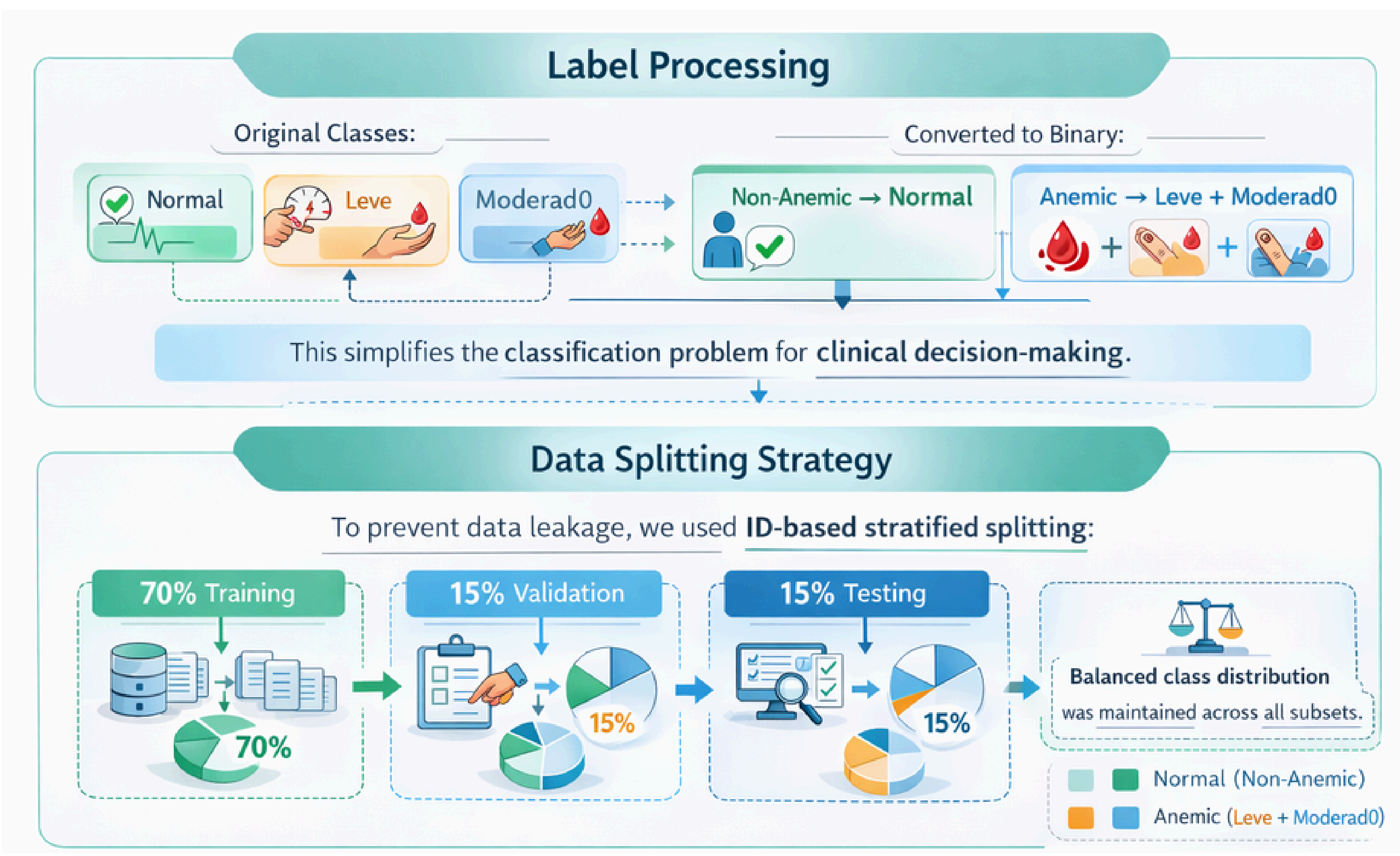
Complex Engineering Problem



Dataset



Data Processing



Baseline Model

Baseline Model Architecture:
Three Unimodel:
- Unas(Nail images)
- Palmas(Plam videos)
- Yemas(Fingertips videos)

Unas Unimodel

Image Verification

Manually verified 742 images (Normal: 519, Mild: 164, Moderate: 59).

Palmas Unimodel

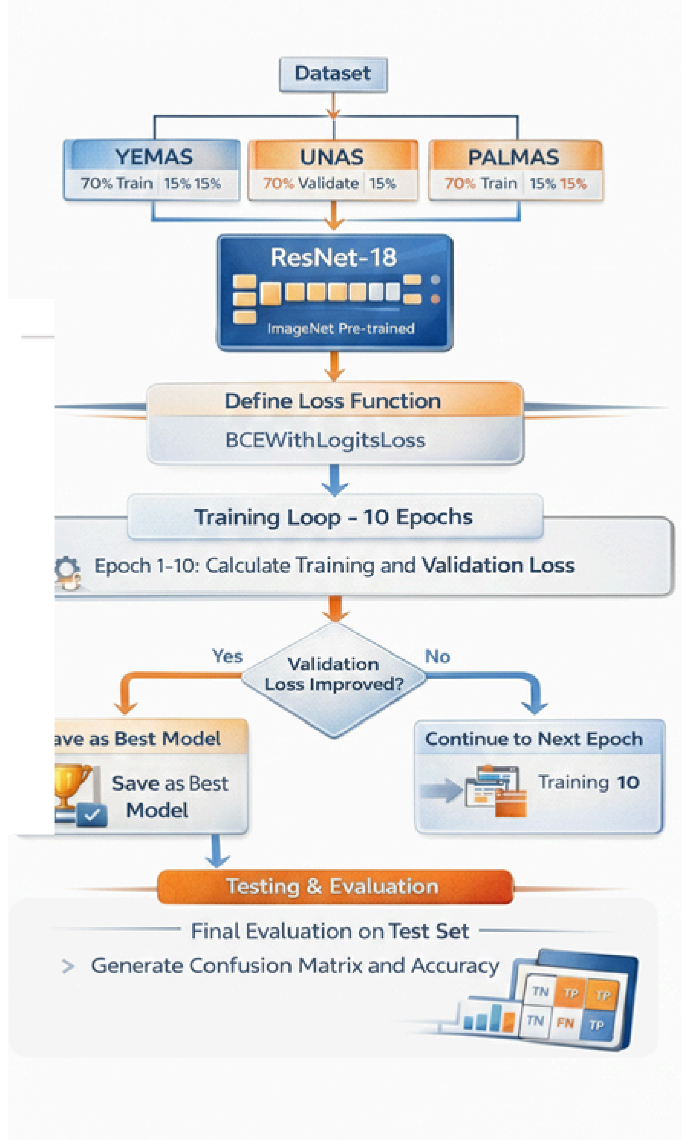
Video Verification & Frame Extraction

Verified 829 videos (Normal: 554, Mild: 201, Moderate: 74). 16 frames sampled per video, resized to 224x224.

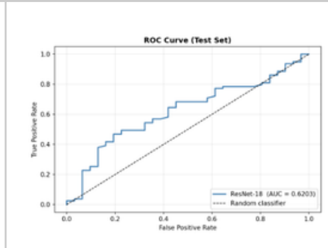
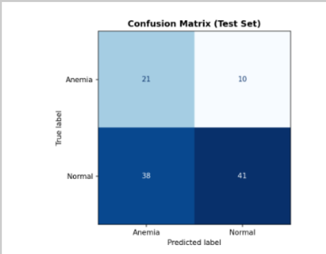
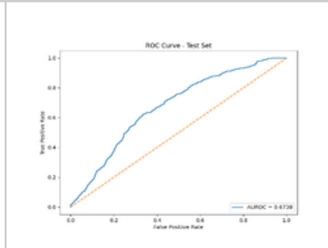
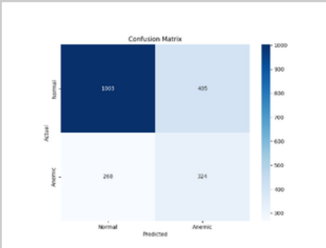
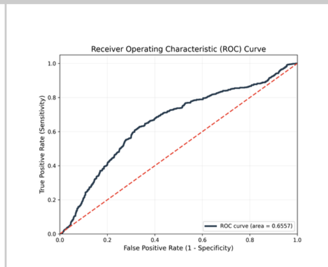
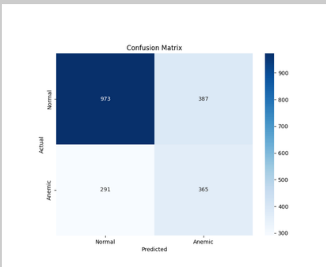
Yemas Unimodel

Video Verification

834 videos manually verified (Mild: 201, Moderate: 75, Normal: 558).



Result

Accuracy	F-1 score	Specificity	Sensitivity	Roc curve	Confusion matrix
Unas					
AUC-ROC : 0.6203 Accuracy : 0.5636	0.6308	0.6774	0.519		
Palmas					
AUC-ROC : 0.673 Accuracy : 0.663	0.49	0.71	0.519		
Yemas					
AUC-ROC : 0.6557 Accuracy : 0.6637	0.5185	0.7154	0.5564		

Future Work

- Building a multimodal model that combines **UNAS(nail)**, **PALMAS(palm)**, and **YEMAS(fingertip)** images.
- Developing a real-time mobile/web screening application.
- Validating the system clinically by comparing it with laboratory hemoglobin tests.

Reference

[1] World Health Organization. Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity. Technical report, WHO, Geneva, Switzerland, 2011.
[2] Nahid Salma and Majid Khan Majahar Ali. Evaluating the factors and forecast-ing childhood anemia through machine learning algorithms. Malaysian Journal of Fundamental and Applied Sciences, 21(1):1529–1541, 2024.
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[4] Subham Das and Mitradip Bhattacharjee. Image-based sensing of leukonychia for early diagnosis of anemia using a smartphone application. IEEE Sensors Letters, 6(5):1–4, 2022.